

Kasra Ferdowsi

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Education

UC San Diego - PhD in Computer Science - M.S. in Computer Science - Advised by Nadia Polikarpova and Sorin Lerner	2019 - 2024
UC Berkeley - B.S. in Electrical Engineering and Computer Science - Graduated with Highest Honors	2017 - 2019
Diablo Valley College - A.S. for Transfer in Mathematics - A.S. for Transfer in Physics	2014 - 2017

Work Experience

Staff Software Engineer <i>Cubist Inc</i> - Building robust and secure software with Rust and TypeScript.	2025 - Present
Senior Software Engineer <i>Cubist Inc</i> - Collaborated on design and development for Confidential Cloud Functions. - Implemented first-class policies to meet customer needs. - Maintained, improved, and extended various features across Cubist products.	2024 - 2025
Engineering Intern <i>Tom Sawyer Software</i> - Designed and developed the ability to deploy <i>Tom Sawyer Perspectives</i> applications to Kubernetes. - Lead the effort to document and improve <i>Tom Sawyer Perspectives</i>' deployment user experience.	Summer 2021
Engineering Intern <i>Tom Sawyer Software</i> - Co-designed and developed a tool using Git, Maven and Docker to streamline building and deploying <i>Tom Sawyer Perspectives</i> applications. - Main contributions included user-interface additions and improvements, an overhaul of the version control system, and bug fixes and improvements.	Summer 2020
Engineering Intern <i>Tom Sawyer Software</i> - Helped design and develop the <i>Tom Sawyer Licensing Server 2.0</i>, a backwards-compatible scalable web server using a microservice architecture. - Implemented secure and user-friendly account management features, and a responsive web interface for <i>Graph Database Browser</i>, as well as database organization, query improvements, and other fixes.	2016 - 2019

Publications

Human-Centered Program Synthesis Ferdowsifard, K.	Dissertation
Bottom-up Synthesis of Memory Mutations with Separation Logic Ferdowsi, K., Peleg, H.	ECOOP 2025
Validating AI-Generated Code with Live Programming Ferdowsi, K., Huang, R., James, M. B., Polikarpova, N., and Lerner, S.	CHI 2024

Publications (Cont.)

ColDeco: An End User Spreadsheet Inspection Tool for AI-Generated Code Ferdowsi, K., Williams, J., Drosos, I., Gordon, A.D., Negreanu, C., Polikarpova, N., Sarkar, A., and Zorn, B.	VL/HCC 2023
Investigating the Impact of Using a Live Programming Environment in a CS1 Course Huang, R., Ferdowsi, K., Selvaraj, A., Soosai Raj, A.G., and Lerner, S.	SIGCSE TS 2022
LooPy: Interactive Program Synthesis with Control Structures Ferdowsifard, K., Barke, S., Peleg, H., Lerner, S., and Polikarpova, N.	OOPSLA 2021
Small-Step Live Programming by Example Ferdowsifard, K., Ordookhanians, A., Peleg, H., Lerner, S., and Polikarpova, N.	UIST 2020

Research Experience

Bottom-up Synthesis of Side-Effects with Separation Logic - With Hila Peleg. - Reformulated bottom-up enumerative synthesis as proof search in Concrete Heap Separation Logic, allowing for efficient and provably correct synthesis in the presence of mutating components. - Implemented and evaluated the technique, showing that it is comparable to the state-of-the-art while offering determinism and a correctness guarantee. - Link to publication: https://doi.org/10.4230/DARTS.11.2.2	2024
LEAP: Validating AI-Generated Code with Live Programming - With Ruanqianqian (Lisa) Huang, Michael B. James, Sorin Lerner, and Nadia Polikarpova. - Designed and implemented a novel interaction model for AI programming assistants that support exploring multiple code suggestions through Live Programming. - Ran a 17 participant between-subjects user study showing numerous benefits to this interaction model, including preventing over-reliance, and reducing cognitive load. - Link to publication: https://doi.org/10.1145/3613904.3642495	2024
Towards Human-Centered Types and Type Debugging - PLATEAU'23 Workshop paper and presentation. - Surveyed the history of the usability of type systems, and type error messages. And offered concrete future directions based on cutting-edge HCI research. - Link to publication: https://doi.org/10.1184/R1/22227457.v1	2022
ColDeco: An End-User Spreadsheet Inspection Tool for AI-Generated Code - With Jack Williams, Ian Drosos, Andrew Gordon, Carina Negreanu, Advait Sarkar, and Ben Zorn. - Designed and implemented a novel interaction model for inspecting and diagnosing errors in AI-generated code, targeted at end-users. - Conducted a 24-participant controlled experiment to evaluate the interaction. - Link to the paper: https://ieeexplore.ieee.org/document/10305647	2022
The Usability of Advanced Type Systems - PhD Research Exam. - Surveyed publications on advanced type systems which inspired the Rust programming language, as well as all usability studies on Rust itself. - Link to the written report: https://arxiv.org/abs/2301.02308	2022
LooPy: Interactive Program Synthesis with Control Structures - With Shraddha Barke, Hila Peleg, Sorin Lerner, and Nadia Polikarpova. - Designed and implemented a novel technique for using Bottom-up Enumerative Synthesis for efficient synthesis of blocks of Python code. - Link to publication: https://dl.acm.org/doi/10.1145/3485530	2020 - 2021

Research Experience (Cont.)

Impact of Live Programming in a CS1 Course	2020 - 2021
- With Ruanqianqian Huang, Ana Selvaraj, Adalbert Gerald Soosai Raj and Sorin Lerner. - Implemented a web editor for the Live Programming environment <i>Projection Boxes</i>, and developed and deployed a server for collecting usage data from the editor. - Ran a user study of hundreds of students in a CS1 course, and visualized and performed statistical analysis on data collected from the study. - Link to the paper: https://dl.acm.org/doi/10.1145/3478431.3499305	
SnipPy: Small-Step Live Programming by Example	2020
- With Allen Ordookhanians, Hila Peleg, Sorin Lerner, and Nadia Polikarpova. - Designed and implemented a novel Program Synthesis interaction model with <i>Projection Boxes</i>. - Ran a 12-participant user study for evaluating the interaction. - Link to publication: https://dl.acm.org/doi/10.1145/3379337.3415869	

Teaching Experience

Teaching Assistant CSE130: Programming Languages	Fall 2023
- Presented discussion sections, held office hours, graded assignments, and assisted students on Piazza.	
Teaching Assistant CSE230: Graduate Programming Languages	Fall 2022
- Presented discussion sections, held office hours, graded assignments, and assisted students on Piazza.	
Teaching Assistant CSE130: Programming Languages	Winter 2022
- Presented discussion sections, held office hours, and assisted students on Piazza.	
Teaching Assistant CSE130: Programming Languages	Fall 2021
- Presented discussion sections, held weekly office hours, and assisted students on Piazza.	
Teaching Assistant CSE11: Accelerated Intro to Programming	Spring 2020
- Took responsibility for one week of the course in the instructor's absence, leading 2 lectures (in a live Q&A format with pre-recorded videos), and giving an online midterm exam to over 200 students. - Led the weekly discussion sections, including writing review notes and practice problems tailored to the course, leading live discussions, and creating pre-recorded videos.	
Teaching Assistant CSE131: Compiler Construction	Fall 2019
- Created autograder tests and infrastructure for the class projects. - Assisted in the design of course assignments. - Updated the compilation target from Intel x86 to x86_64.	

Service

Student Organizer	PLATEAU 2024
External Reviewer	CHI 2023
Program Committee Member	HATRA 2023
Student Volunteer	PLDI 2023

Mentorship

GradWIC Mentor	2022 - 2023
CSE599: Teaching Methods in Computer Science Mentor	Winter 2023
ERSP Mentor	2021 - 2022
- With Ruanqianqian (Lisa) Huang, Nadia Polikarpova and Sorin Lerner - Link to poster: https://drive.google.com/file/d/1HZx_FODLaYjbIDEKttTFO5jsF8EUov_Y/	